

SHAHZAIN ALI

AI Automation Developer

Karachi, Pakistan · Open to Remote (UTC+5) · +92 347 3890095 · enshahzainali110@gmail.com
github.com/Shahzain-Ali · linkedin.com/in/shahzain-ali1 · shahzainaliportfolio-website-vert.vercel.app

SUMMARY

AI Automation Developer building production agentic systems: multi-agent pipelines (OpenAI Agents SDK, Google ADK), RAG applications with vector databases (Qdrant), MCP server development, and business workflow automation (n8n, WhatsApp Cloud API). Shipped an autonomous AI Employee (7 MCP servers, 31 tools) and a RAG-powered documentation assistant with cited answers. Strong on human-in-the-loop design, FastAPI backends, and Docker deployment.

PROJECTS

WhatsApp AI Receptionist — Google ADK · github.com/Shahzain-Ali/whatsapp-ai-receptionist

- 24/7 clinic receptionist on WhatsApp: answers strictly from a grounded knowledge base and books appointments end-to-end with zero staff involvement, against real-time Google Sheets availability
- Google ADK LlmAgent (OpenAI GPT-4o-mini via LiteLLM) with tool calling, a before_tool_callback double-booking guardrail, optional WhatsApp human-approval mode, and full trace/span observability (adk web)
- **Stack:** Google ADK · FastAPI · WhatsApp Cloud API · Docker · deployed live on Render

AI Employee System (Digital FTE) · github.com/Shahzain-Ali/ai-employee-fte-system

- Autonomous AI Employee managing email, social media (5 platforms), and accounting workflows 24/7, with human-in-the-loop approval for sensitive actions
- Architected 7 MCP servers exposing 31 tools and 17 agent skills; Perception → Reasoning → Action pipeline on the Claude Code agent runtime
- **Stack:** Python · MCP · Claude (Anthropic) · PostgreSQL · Playwright · Streamlit · Docker

AI Tutor — Agentic Solutions Interactive Book (Agentic RAG) · shahzain-ali.github.io/agentic-solutions-book (live)

- Live AI tutor embedded in a multi-book learning platform: agentic RAG where retrieval is a tool the agent calls via OpenAI Agents SDK function calling, answering strictly from book content with relevance-filtered citations
- Qdrant Cloud vector search (OpenAI text-embedding-3-small) with a heading-aware chunking pipeline; production hardening: per-IP rate limiting, budget caps, CORS, input sanitization; chat history in Neon Postgres
- **Stack:** OpenAI Agents SDK · GPT-4o-mini · Qdrant · FastAPI · Docker · Render · GitHub Actions CI/CD

AutoInvoice AI — Agentic Invoice Generator · github.com/Shahzain-Ali/agentic-invoice-generator

- 3-agent pipeline (validation → template insertion → delivery) that turns Google Sheets / Excel / web-form data into delivered PDF invoices in under 60 seconds, zero manual steps
- **Stack:** OpenAI Agents SDK · GPT-4o-mini · FastAPI · n8n · Streamlit · Jinja2

SKILLS

- **LLMs & Prompting:** OpenAI (GPT-4o / 4o-mini), Google Gemini, Anthropic Claude, function calling, structured output, model routing (LiteLLM), prompt engineering
- **Retrieval & Vector:** RAG, Qdrant, embeddings (OpenAI text-embedding-3-small), semantic search
- **Agent Frameworks:** Google ADK, OpenAI Agents SDK, MCP (Model Context Protocol) server development
- **Automation & Integration:** n8n, WhatsApp Cloud API, webhooks, REST APIs, human-in-the-loop design
- **Backend & Data:** Python (async), FastAPI, Pydantic, PostgreSQL
- **Observability & Tracing:** OpenAI Traces dashboard, Google ADK trace/span (adk web)
- **Deployment & DevOps:** Docker, GitHub Actions (CI/CD), Render, Hugging Face Spaces, Vercel
- **Tools:** Git & GitHub, Claude Code, Streamlit, Spec-Driven Development

EDUCATION

- **BS Computer Science** — Virtual University of Pakistan · Expected 2028
- **Agentic AI Architect** — Governor House GIAIC, Karachi · In Progress

CERTIFICATIONS

- Introduction to Model Context Protocol — Anthropic
- n8n: No-Code AI Agent Builder — Simplilearn

Languages: English (professional), Urdu (native)